

ENTROPICITY: A Unified Theory of Reality as Entropic Computation

*Foundations, Mathematical Structure, and Philosophical Implications
of the Theory of Entropicity (ToE)*

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Abstract

We present the **Theory of Entropicity (ToE)** — a complete, unified theoretical framework in which entropy is not a derived thermodynamic quantity but the fundamental ontological substrate of physical reality. In the Theory of Entropicity, all known physics — spacetime geometry, quantum field theory, gravity, matter, and causality — emerges from the flow and gradient of entropy on an underlying **entropic manifold** $\mathbb{N}$ equipped with a Fisher-Rao information metric. The central variational object of the theory is the **Obidi Action**, a generalized action functional defined on $\mathbb{N}$ that subsumes the Einstein-Hilbert action, the Standard Model action, and quantum information theory as limiting cases. Variation of the Obidi Action

with respect to the entropic metric yields the **Master Entropic Equation** — a unified field equation generalizing Einstein's field equations to include entropy-stress and information-density tensors on the right-hand side, and entropic curvature corrections on the left. We derive the **Entropic Einstein Equation** as the classical limit of the Master Entropic Equation, recovering standard General Relativity when entropy is at equilibrium. We establish and prove the **No-Rush Theorem** (Misiobidi, 2026), which states that the rate of entropic change along any physical trajectory is bounded above by a maximum entropic velocity proportional to the Margolus-Levitin quantum speed limit. This theorem unifies the second law of thermodynamics, the quantum speed limit, and the emergence of time into a single geometric statement on $\mathbb{N}$. We interpret the universe as a **self-computing entropic system**, extending Wheeler's "It from Bit" and Wolfram's Ruliad to the entropic domain: physical laws are the fixed points of the entropic feedback loop, and the Big Bang is a spontaneous symmetry-breaking instability of the entropic vacuum. Predictions include modified dispersion relations testable with gamma-ray bursts, Tsallis-distributed observables in quantum condensed-matter systems, and gravitational wave echoes near black hole horizons. The Theory of Entropicity provides a unified, mathematically rigorous, and philosophically coherent foundation for all of physics with entropy as its ontological primitive.

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1. Introduction: The Entropic Paradigm Shift

Physics has, over the last century, undergone two revolutionary restructurings of its conceptual foundations. The first was the overthrow of Newtonian absolute space and time by Einstein's special and general theories of relativity, in which geometry became a dynamical entity rather than a passive stage. The second was the quantum revolution, in which the deterministic clockwork of classical mechanics was replaced by probability amplitudes, wave functions, and the irreducible randomness of measurement. Both revolutions, however, retained a common implicit assumption: that energy — and the matter and fields that carry it — constitutes the fundamental currency of physical reality. The **Theory of Entropicity (ToE)**, presented in this monograph, proposes a third revolution: that entropy, not energy, not matter, and not spacetime geometry, is the fundamental ontological primitive from which all of physics is derived.

The conceptual seeds of this revolution were planted by John Archibald Wheeler, who in 1990 introduced the slogan "It from Bit," arguing that every physical entity derives its existence and meaning from the answers to binary yes-or-no questions — from information [1]. This information-theoretic turn was radical in its implications: if physical reality is ultimately informational, then the laws of physics are, at bottom, laws about the processing, encoding, and transformation of information. Wheeler's program was visionary but remained incomplete: he provided no specific mathematical mechanism by which information could generate the specific structures of general relativity, quantum field theory, or the Standard Model.

In parallel, Ludwig Boltzmann's definition of thermodynamic entropy as $S = k_B \ln \Omega$ — where Ω is the number of accessible microstates and k_B is Boltzmann's constant — established entropy as a central quantity in physics [3]. However, in the Boltzmann-Gibbs-Shannon tradition, entropy has always been understood as a *derived* quantity: it is computed from an underlying probability distribution over microstates, which are themselves defined by the dynamical laws of the theory. Entropy, in this picture, emerges from the underlying physics; it does not generate it.

The Theory of Entropicity inverts this ontological hierarchy. We postulate that entropy is not emergent but **foundational**: the dynamics, geometry, and causal structure of physical reality are expressions of entropic flow on a fundamental information-geometric structure we call the **entropic manifold** $\mathbb{N}$. In this framework, spacetime is a coarse-grained projection of $\mathbb{N}$; quantum fields are entropic fluctuations of $\mathbb{N}$ around its ground state; gravity is the geometric curvature of $\mathbb{N}$ sourced by entropy and information gradients; and the laws of nature are geodesic equations on $\mathbb{N}$.

The motivation for this radical inversion is not merely philosophical but empirical. Several convergent lines of inquiry in modern theoretical physics point toward the primacy of entropy:

- Jacobson (1995) derived the full Einstein field equations from the thermodynamics of local Rindler horizons, treating the equations of motion as an equation of state [13].
- Verlinde (2011, 2016) showed that Newtonian gravity, and subsequently dark energy, can be understood as entropic forces arising from the gradient of entropy on holographic screens [15, 16].
- The Bekenstein-Hawking entropy formula $S_{\text{BH}} = A k_{\text{B}} c^3 / (4 G \hbar)$ establishes that black holes carry an entropy proportional to their surface area, suggesting that information geometry is encoded on boundaries [9, 10].
- The Maldacena AdS/CFT correspondence establishes a precise duality between gravitational physics in the bulk and conformal quantum field theory on the boundary, with entropy as the bridge quantity [26].
- Wolfram's computational universe program (2020) identifies the evolution of the universe with the maximal entangled limit of all computational rules — the Ruliad — suggesting that physical laws are computational attractors [28].

The Theory of Entropicity synthesizes and supersedes all of these programs. It does so through three principal mathematical constructions: the **Obidi Action**, a master variational functional defined on the entropic manifold; the **Master Entropic**

Equation, the unified field equation obtained by varying the Obidi Action; and the **No-Rush Theorem**, a fundamental bound on the rate of entropic evolution that unifies the quantum speed limit, the second law of thermodynamics, and the emergence of time as a single geometric statement.

The paper is organized as follows. Section 2 introduces the conceptual foundations of Entropicity, including the formal definition of the theory and its philosophical lineage. Section 3 develops the mathematical machinery of the entropic manifold, including information geometry, non-extensive entropies, and spectral algebra of entropic operators. Section 4 states and motivates the Obidi Action in full. Section 5 derives and interprets the Master Entropic Equation. Section 6 derives the Entropic Einstein Equation as the classical limit. Section 7 states and proves the No-Rush Theorem. Section 8 develops the computational universe interpretation. Section 9 connects ToE to quantum field theory. Section 10 provides a systematic comparative analysis with existing theories. Section 11 derives experimental predictions. Section 12 analyzes the entropic feedback loop in detail. Section 13 explores philosophical implications. Section 14 concludes. Appendices A-E provide technical derivations and reference tables.

2. Conceptual Foundations of Entropicity

2.1 What Is Entropicity?

We introduce **Entropicity** as a property of physical systems, defined as follows:

Definition (Entropicity)

A physical system possesses **Entropicity** if and only if its dynamics, geometry, and causal structure are entirely and completely determined by the flow, gradient, and curvature of entropy on the entropic manifold $\mathbb{N}$ associated with that system. A theory possesses Entropicity if and only if all of its equations of

motion are derivable as geodesic, extremal, or conservation equations on $\mathbb{N}$.

Entropicity must be carefully distinguished from thermodynamics, statistical mechanics, and Shannon-type information theory, with which it shares mathematical tools but differs fundamentally in ontological status. In classical thermodynamics, entropy is a macroscopic state function derived from the underlying microdynamics. In statistical mechanics, entropy is a measure of ignorance or coarse-graining over microstates. In information theory, entropy is a measure of uncertainty over probability distributions. In all three frameworks, entropy is a *secondary* quantity computed from a substrate of more fundamental objects (microstates, fields, or probability distributions).

In the Theory of Entropicity, this relationship is reversed. **Entropy is the ontological primitive**. Microstates, probability distributions, fields, and spacetime geometry are all *constructions* — projections, representations, or coarse-grainings of the entropic manifold $\mathbb{N}$. The entropic manifold is not a space over which entropy is defined; it is a space that *is* entropy, in the same sense that Riemannian geometry is curvature. Entropy, in ToE, is not a function on a space; it is a structural property of the space itself. This distinction — between entropy as a derived epistemic quantity and entropy as an ontological primitive — is the axiomatic foundation of the entire theory.

More precisely, in ToE the entropy S of a region of the entropic manifold is not computed from an underlying probability distribution; rather, the probability distribution is defined as the gradient of S with respect to the entropic coordinates. The dynamics of a physical system are not governed by a Hamiltonian that one then uses to compute S ; instead, S itself generates the dynamics through the entropic geodesic equation. This ontological primacy of entropy, stated formally, constitutes what we call the **Entropicity Axiom**: entropy precedes dynamics.

2.2 The Entropic Substrate Hypothesis

The central structural postulate of the Theory of Entropicity is the **Entropic Substrate Hypothesis**, which we state as follows:

Postulate (Entropic Substrate Hypothesis)

The physical spacetime manifold M (the arena of general relativity and quantum field theory) is a coarse-grained projection of a more fundamental structure, the **entropic manifold** N . The entropic manifold is a smooth, complete Riemannian manifold whose coordinates $\{q^i\}$ parametrize probability distributions over physical microstates. The metric on N is the Fisher-Rao information metric $g^{(\varepsilon)}_{\mu\nu}$ derived from the information geometry of those probability distributions. Every point $p \in M$ in physical spacetime corresponds to an equivalence class of probability distributions $[\rho_p]$ over microstates, and the geometry of M at p is determined by the entropic geometry of N at the corresponding point.

This postulate has several immediate and non-trivial consequences. First, the metric structure of spacetime is not independent of the matter and entropy content of spacetime — a fact already embodied in general relativity — but in ToE this dependence is deeper: the metric is literally *constituted by* information-geometric relationships between probability distributions. The curvature of spacetime is the curvature of information space.

Second, every point in spacetime carries a non-trivial entropic structure. This means that spacetime is not merely a container for matter and fields; it is a structure with an irreducible informational content at every point. This informational content is what we identify with the vacuum energy, the zero-point fluctuations of quantum field theory, and ultimately with the observed cosmological constant.

Third, the distinction between "classical" and "quantum" physics is, in ToE, a distinction between different regimes of the entropic manifold — not a distinction between fundamentally different theories. The quantum regime corresponds to regions of N where the Fisher metric is highly curved (high information density), while the classical regime corresponds to flat regions of N (low information curvature). The transition between quantum and classical behavior is, in ToE, a phase transition on N .

2.3 Philosophical Lineage

The Theory of Entropicity does not arise in a philosophical vacuum. It stands in explicit dialogue with three major intellectual traditions in the foundations of physics:

Wheeler's "It from Bit" (1990). Wheeler proposed that every physical entity derives its existence from informational acts of observation [1]. In ToE, we make this precise: bits are the $q \rightarrow 1$ (extensive) limit of entropic information units; the universe is not merely informational but *entropic-computational*. Wheeler's participatory universe — in which observers play a constitutive role in creating reality — is rendered mathematically precise in ToE as the statement that observation constitutes an entropic collapse: the act of measurement selects a definite microstate from the entropic superposition, reducing the entropy of the observed subsystem while increasing the entropy of the observer-environment system. The universe is participatory because every physical interaction is an entropic computation, and every computation modifies the entropic manifold.

Wolfram's Ruliad (2020). Wolfram's program identifies the universe with the maximal entangled limit of all possible computational rules — a structure he calls the Ruliad [28]. In ToE, we provide a natural identification: the Ruliad is the entropic manifold $\mathbb{N}$. Different "rules" in Wolfram's sense correspond to different regions of $\mathbb{N}$ with different entropic curvatures. Computational irreducibility — the impossibility of predicting the evolution of certain computational systems without running them step by step — corresponds in ToE to regions of maximal entropic curvature: areas where the geodesic equations on $\mathbb{N}$ cannot be integrated in closed form. Wolfram's "observer threads" correspond to entropic sub-manifolds of $\mathbb{N}$ — coherent, persistent sub-structures that sample consistent slices of the entropic superposition.

Verlinde's Entropic Gravity (2011, 2016). Verlinde derived Newtonian gravity as an entropic force $F = T(\Delta S/\Delta x)$ arising from the gradient of entropy on a holographic screen, and later proposed that dark energy and dark matter are manifestations of the entropy associated with the de Sitter horizon [15, 16]. ToE subsumes and extends Verlinde's program: in the language of the Master Entropic Equation, Verlinde's entropic force is the weak-field, non-relativistic limit of the entropy-stress

tensor $T^{(\text{entropy})}_{\mu\nu}$. ToE extends Verlinde by (a) making the theory fully covariant through the Obidi Action, (b) incorporating non-extensive Tsallis entropy to handle long-range correlations and the gravitational self-interaction, (c) providing a natural quantum extension via the spectral algebra of entropic operators, and (d) explaining the origin of the holographic screen as a boundary of the entropic manifold.

3. Mathematical Foundations

3.1 The Entropic Manifold $\mathbb{N}$

We define the **entropic manifold** $\mathbb{N}$ as a smooth, complete, finite-dimensional Riemannian manifold with local coordinates $\{q^i\}$, $i = 1, \dots, n$, where each point $q \in \mathbb{N}$ represents a smooth probability distribution $p(x|q)$ over a measurable space X . The manifold $\mathbb{N}$ is equipped with the **Fisher-Rao metric**:

$$\begin{aligned}
 (3.1) \quad & g \\
 (F) \quad & ij \\
 (q) = & \\
 \int & \\
 p(x|q) [& \\
 \partial & \\
 i & \\
 \ln p(x|q)] [& \\
 \partial &
 \end{aligned}$$

j

$\ln p(x|q)] dx$

where $\partial_i \equiv \partial/\partial q^i$. The Fisher-Rao metric is the unique (up to a constant factor) Riemannian metric on the space of probability distributions that is invariant under sufficient statistics — a result known as the Chentsov theorem [17]. This uniqueness is a crucial motivation for the choice of the Fisher-Rao metric in ToE: it is the only canonically defined information-geometric structure available.

The geodesic distance on $\mathbb{N}$ between two probability distributions p and p' is:

(3.2)

D

F

$(p, p') = 2 \arccos [$

$\int$

$\sqrt$

$p(x)$

$\sqrt$

$p'(x) dx] = 2 \arccos[B(p, p')]$

where $B(p, p')$ is the Bhattacharyya coefficient. This geodesic distance — the **Bhattacharyya angle** — provides the fundamental measure of distinguishability between physical states in ToE. The quantum version of this distance is the Bures metric, which reduces to the Fisher-Rao metric for pure states.

A central theorem of information geometry establishes that the entropic manifold $\mathbb{N}$ simultaneously generalizes both the arena of general relativity and the arena of quantum mechanics:

- **GR limit:** When the probability distributions $p(x|q)$ are sharply peaked (low entropy, classical regime), the Fisher-Rao metric $g^{(F)}_{ij}$ reduces to the Riemannian metric of a classical configuration space, and geodesics on $\mathbb{N}$ reduce to classical trajectories in spacetime.
- **QM limit:** When $p(x|q)$ are arbitrary mixed states (density matrices ρ), the Fisher manifold $\mathbb{N}$ is isomorphic to the space of density matrices on a Hilbert space, and the geometry of $\mathbb{N}$ reproduces the geometry of quantum state space.

Geodesics on $\mathbb{N}$ — curves $\gamma(\tau)$ that minimize the Fisher-Rao length — represent **paths of minimum entropic change**. In ToE, these geodesics are the physical trajectories of all systems: they are what classical mechanics calls "least-action paths" and what quantum mechanics calls "dominant paths in the path integral." The geodesic principle is thus the entropic unification of the principle of least action and the principle of stationary phase.

3.2 Non-Extensive Entropies and the Tsallis-Rényi Framework

Standard Boltzmann-Gibbs statistical mechanics rests on the additive (extensive) entropy:

(3.3)

S

BG

$=$

$-$

k

B

Σ

i

p

i

$\ln p$

i

Extensivity means that for two independent systems A and B: $S(A \cup B) = S(A) + S(B)$. This property holds for systems with short-range interactions and negligible long-range correlations. However, it fails precisely for the systems most relevant to quantum gravity: black holes, cosmological spacetimes, self-gravitating systems, and systems with long-range entropic correlations. For these systems, the **Tsallis q-entropy** provides the appropriate generalization [6]:

(3.4)

S

q

= k

B

$\cdot (1$

–

Σ

i

p_i^q
 $\sum_i p_i^q$
 $1/q$
 $1)$
 $(q \neq 1)$

with the limit $\lim_{q \rightarrow 1} S_q = S_{BG}$ recovered by L'Hôpital's rule. The parameter $q \in \mathbb{R}$ is the **non-extensivity parameter**, characterizing the degree of correlation in the system. For $q > 1$, sub-additivity holds: $S_q(A \cup B) \leq S_q(A) + S_q(B)$, appropriate for systems with positive correlations (such as gravitationally bound systems). For $q < 1$, super-additivity holds, appropriate for anti-correlated systems.

The **Rényi entropy** provides a complementary generalization [7]:

(3.5)
 S_α
 $= (1/(\alpha - 1)) \ln$

Σ

i

p

i

α

)

(

α

$\neq$

1,

α

$>$

0)

also recovering S_{BG} in the limit $\alpha \rightarrow 1$. In ToE, we use the Tsallis entropy as the primary entropy measure because it arises naturally from the Obidi Action through variation, while the Rényi entropy provides the correct framework for the spectral entropy of the entropic operator. A crucial result connecting non-extensivity to spacetime geometry is established by the following correspondence:

Theorem (Curvature-Extensivity Correspondence)

The Ricci curvature scalar $\mathfrak{R}_\epsilon$ of the entropic manifold $\mathbb{N}$ at a point q corresponds to the second-order deviation of the Tsallis parameter q from unity:

(3.6)

R

$\mu\nu$

(q)

$\approx$

∂

μ

∂

ν

$(q$

$-$

$1) + O((q$

$-$

$1)^2)$

where $R_{\mu\nu}(q)$ is the entropic Ricci tensor evaluated at the coordinate q on $\mathbb{N}$. This identifies spacetime curvature with the non-extensivity of the underlying entropy.

This theorem has profound consequences: it means that gravitational curvature is not an independent geometric structure imposed on spacetime, but is a manifestation of the degree of non-extensivity of the entropy of the physical system. Where gravity is strong (high curvature), entropy is strongly non-extensive (q significantly different from 1). In flat spacetime, $q \rightarrow 1$ and extensivity is recovered.

3.3 Spectral Algebra of Entropic Operators

To quantize the Theory of Entropicity, we introduce a Hilbert space formalism on $\mathbb{N}$. Let $\mathcal{H}$ be the Hilbert space of square-integrable functions on $\mathbb{N}$, and let ρ be a density matrix (positive semi-definite, trace-one operator) on $\mathcal{H}$ representing the quantum state of a physical system. We define the **Entropic Operator** $\hat{E}$ as:

(3.7)

$\hat{E}[\rho]$

ρ

$] =$

$-$

k

B

$\text{Tr}(\rho \ln \rho)$

ρ

$\ln$

ρ

$) = S_{\text{vN}}$

vN

$[\rho]$

ρ

$]$

where S_{vN} is the Von Neumann entropy. The spectral decomposition of the density matrix ρ is:

(3.8)

ρ

$=$

$\sum$

i

λ

i

$|$

ψ

i

$\rangle$

$\langle$

ψ

i

$|$

where $\{|\psi_i\rangle\}$ are the eigenstates of ρ and $\{\lambda_i\}$ are the corresponding eigenvalues, satisfying $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$. The eigenvalues λ_i encode the **entropic weights** of the corresponding pure states: the degree to which each basis state contributes to the total entropy of the system. The **Entropic Spectrum** $\{\lambda_i\}$ determines the complete informational content of a region of the entropic manifold — analogous to the way the energy spectrum $\{E_n\}$ determines the quantum mechanical properties of a system.

We now introduce the fundamental algebraic structure of the quantized ToE. Define the **Entropic Commutator** of $\hat{E}$ with the Hamiltonian operator $\hat{H}$:

(3.9)

$$[\hat{E}, \hat{H}] = i\hbar \hat{S}$$

where $\hat{S}$ is the **entropy-flow operator**, the generator of entropic time translations on $\mathbb{N}$. Equation (3.9) is the fundamental quantum commutation relation of ToE: it expresses the fact that entropy and energy are canonically conjugate in the entropic sense — a statement that will yield the No-Rush Theorem in Section 7. The Von Neumann equation of motion for the density matrix:

(3.10)

$i\hbar \frac{d}{dt}$

ρ

$\frac{d}{dt} \rho = [\hat{H},$

ρ

$] =$

$-$

$i\hbar \hat{S}[\rho]$

ρ

$]$

shows that the rate of change of the density matrix is generated by the entropy-flow operator, establishing that quantum dynamics is entropic dynamics. The **Heisenberg uncertainty principle** emerges as a consequence of the non-commutativity (3.9):

(3.11)

Δ

$S \cdot$

Δ

E

$\geq$

$\hbar/2 \cdot |$

$\langle$

$\hat{S}$

$\rangle$

$|$

This is the *entropic uncertainty relation* of ToE: a lower bound on the product of entropy uncertainty and energy uncertainty. It reduces to the standard energy-time uncertainty relation $\Delta E \Delta t \geq \hbar/2$ when the entropy-flow operator $\hat{S}$ is identified with the rate of entropic change dS/dt , and time is interpreted as the integral of entropic flow (as established by the No-Rush Theorem).

3.4 Information Geometry and the Entropic Connection

The entropic manifold $\mathbb{N}$ is equipped not only with the Fisher-Rao metric but also with a canonical affine connection — the **entropic connection** — derived from the information-geometric structure of the probability distributions. The entropic affine connection (the α -connection of Amari-Nagaoka [17]) for $\alpha = 0$ (the mixture connection) is:

(3.12)

$$\Gamma_{ijk}(\varepsilon) = - \int \frac{\partial}{\partial \theta^i} p(x|q) \frac{\partial}{\partial \theta^j} \ln p(x|q) \cdot \frac{\partial}{\partial \theta^k} \ln p(x|q) dx$$

This connection is compatible with the Fisher-Rao metric and defines a notion of parallel transport on $\mathbb{N}$ that preserves the information content of a probability distribution. The corresponding entropic curvature tensor is:

$$(3.13) \quad R_{ij}(\varepsilon)$$

$ijkl$

$=g$

(F)

im

$[$

∂

k

Γ

$(\varepsilon)m$

jl

$-$

∂

l

Γ

$(\varepsilon)m$

jk

$+$

Γ

$(\varepsilon)m$

kn

Γ

$(\varepsilon)n$

$j|$

–

Γ

$(\varepsilon)m$

$\ln$

Γ

$(\varepsilon)n$

j_k

$]$

The entropic curvature tensor encodes the degree to which probability distributions in different regions of $\mathbb{N}$ are correlated — and, in the physical interpretation, the degree to which different regions of spacetime interact. The entropic force — the force that a system with entropy gradient exerts on its surroundings — is:

(3.14)

F

μ

$= T$

∂

μ

S

where T is the local temperature and S is the entropy of the system. Equation (3.14) is the *entropic force law* of ToE, generalizing Verlinde's result to a fully covariant,

information-geometric context [15]. It states that any gradient of entropy exerts a physical force proportional to the temperature — gravity, electromagnetism, and the strong and weak nuclear forces are, in this framework, all entropic forces corresponding to different types of entropy gradients on $\mathbb{N}$.

The entropic geodesic equation — the equation of motion for a free particle on $\mathbb{N}$ — is:

(3.15)

d^2q

i

$/d$

τ

$^2 +$

Γ

$(\varepsilon)i$

jk

$(dq$

j

$/d$

τ

$)(dq$

k

$/d$

τ

$$) = 0$$

This is formally identical in structure to the geodesic equation of general relativity, but with the Levi-Civita connection replaced by the entropic connection $\Gamma^{(\epsilon)}$. In the GR limit (where $\mathbb{N}$ reduces to a Riemannian spacetime), this reduces exactly to the relativistic geodesic equation of a free-falling particle in curved spacetime.

4. The Obidi Action

4.1 Derivation and Motivation

The central variational object of the Theory of Entropicity is the **Obidi Action** — named in honor of the theoretical insight that entropy, not energy, is the fundamental variational quantity of nature. The motivation for the Obidi Action proceeds from the following observation: every known action principle in physics — the Einstein-Hilbert action of general relativity, the Yang-Mills action of gauge theories, the Dirac action for fermions — can be understood as specifying the entropy production rate of the corresponding physical system, if one adopts the correct entropic interpretation of the variational principle.

Specifically:

- The Einstein-Hilbert action $S_{\text{EH}} = \int R \sqrt{-g} \, d^4x / (16\pi G)$ is, in ToE, the leading-order approximation to the entropic curvature term when the entropy of spacetime is nearly extensive ($q \approx 1$).
- The Standard Model action $\sim \int [-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi] \, d^4x$ is, in ToE, the entropic fluctuation action of the gauge and matter fields around the ground state of $\mathbb{N}$.
- The quantum information action $\sim \int \text{Tr}(\rho \ln \rho) \, dV$ is the zero-gravity limit of the Obidi Action.

The Obidi Action is the unique action functional that simultaneously:

1. Is defined on the entropic manifold $\mathbb{N}$ with the Fisher-Rao metric.
2. Reduces to all known action principles in the appropriate limits.
3. Is invariant under the full diffeomorphism group of $\mathbb{N}$ (entropic general covariance).
4. Incorporates non-extensive (Tsallis) entropy to account for long-range correlations.
5. Includes the Fisher information as a kinetic term for the metric itself (entropic kinetic energy of spacetime).

4.2 The Obidi Action — Full Statement

The **Obidi Action** is the following functional on the entropic manifold $\mathbb{N}$:

(4.1)

S

Obidi

=

$\int$

$\mathbb{N}$

[

$\mathfrak{R}$

ε

/(16

π

G

ε

) +

$\mathcal{L}$

matter

(

φ

,

∂

φ

, g

(ε)

) +

$\wedge$

ε

· S

q

(

φ

) +

α

· I

Fisher

$$\begin{aligned}
 & \left(g_{\mu\nu} \right. \\
 & \left. \left(\varepsilon_{\mu\nu} \right) \right. \\
 & \left. \right) + \\
 & \beta \\
 & \cdot \text{Tr} \left(\right. \\
 & \rho \\
 & \ln \\
 & \rho \\
 & \left. \right)] \\
 & \sqrt{ } \\
 & |\det g| \\
 & \left(\varepsilon \right) \\
 & | d^4 q
 \end{aligned}$$

where the individual terms are defined and interpreted as follows:

Symbol	Name	Physical Meaning
$\mathfrak{R}_{\varepsilon}$	Entropic Ricci scalar	Curvature of the Fisher-Rao manifold; encodes the information-geometric curvature of spacetime
G_{ε}	Entropic gravitational constant	$G_{\varepsilon} = G \cdot k_B/c^2$, relates gravitational coupling to entropy; bridges Newtonian G and thermodynamics
$\mathcal{L}_{\text{matter}}$	Matter Lagrangian density	Standard matter Lagrangian for fields φ on the entropic manifold, coupling matter to entropic geometry

Symbol	Name	Physical Meaning
Λ_ϵ	Entropic cosmological constant	Entropy density of the vacuum; determines the vacuum energy and dark energy contribution
$S_q(\varphi)$	Tsallis non-extensive entropy of field φ	Non-extensive entropic content of matter; encodes long-range correlations and self-gravity
$I_{\text{Fisher}}(g^{(\epsilon)})$	Fisher information of the metric	Information content of spacetime geometry itself; kinetic term for the entropic metric
α, β	Coupling constants	Govern the strength of entropy-information-matter interactions; fixed by the q and $\hbar$ of the theory
$\text{Tr}(\rho \ln \rho)$	Von Neumann entropy density	Quantum entropic content of the field; negative of the quantum information term
$\sqrt{ \det g^{(\epsilon)} } d^4q$	Entropic volume form	Invariant integration measure on the entropic manifold $\mathbb{N}$

The Obidi Action has the following key structural features. It is a *scalar* under reparametrizations of the entropic coordinates q^i — it satisfies entropic general covariance. The first term $\mathfrak{R}_\epsilon/(16\pi G_\epsilon)$ is the entropic analogue of the Einstein-Hilbert term: it encodes the kinetic energy of the entropic manifold (its "elasticity" to curvature deformations). The term $\Lambda_\epsilon \cdot S_q(\varphi)$ couples the cosmological term directly to the non-extensive entropy of matter — a crucial departure from standard GR, where the cosmological constant is simply an additive constant with no dynamical origin. In ToE, the cosmological "constant" is dynamically generated by the Tsallis entropy of matter and vacuum fields, explaining both its existence and its small value.

4.3 Variational Principle: Deriving the Master Entropic Equation

The equations of motion of the Theory of Entropicity are obtained by demanding that the Obidi Action be stationary under variations of the fields and the metric. Variation with respect to the entropic metric $g^{(\epsilon)}_{\mu\nu}$ yields:

(4.2)

δ

S

Obidi

/

δ

g

(ε)

$\mu\nu$

$= 0$

Computing this variation term by term (details given in Appendix A):

- $\delta(\mathfrak{R}_\varepsilon \sqrt{g})/\delta g^{\mu\nu} = \sqrt{g} \cdot [R^{(\varepsilon)}_{\mu\nu} - \frac{1}{2}g^{(\varepsilon)}_{\mu\nu}\mathfrak{R}_\varepsilon] = \sqrt{g} \cdot G^{(\varepsilon)}_{\mu\nu}$ — the Entropic Einstein tensor.
- $\delta(\Lambda_\varepsilon S_q \sqrt{g})/\delta g^{\mu\nu} = \sqrt{g} \cdot [\Lambda_\varepsilon g^{(\varepsilon)}_{\mu\nu} S_q + \Lambda_\varepsilon \Phi_{\mu\nu}]$ — the Tsallis correction tensor.
- $\delta(\alpha \mathbf{I}_{\text{Fisher}} \sqrt{g})/\delta g^{\mu\nu} = \sqrt{g} \cdot \alpha T^{(\text{information})}_{\mu\nu}$ — the Fisher information stress tensor.
- $\delta(\beta \text{Tr}(\rho \ln \rho) \sqrt{g})/\delta g^{\mu\nu} = \sqrt{g} \cdot \beta T^{(\text{entropy})}_{\mu\nu}$ — the entropy stress tensor.
- $\delta(\mathcal{L}_{\text{matter}} \sqrt{g})/\delta g^{\mu\nu} = \sqrt{g} \cdot (-\frac{1}{2})T^{(\text{matter})}_{\mu\nu}$ — the standard energy-momentum tensor.

Collecting all terms and setting the total variation to zero yields the **Master Entropic Equation**, stated in full in Section 5. Variation of the Obidi Action with respect to the matter fields φ yields:

(4.3)

δ

S

Obidi

/

δ

φ

$= 0$

$\Rightarrow$

[$\square$

ε

$+ m^2 +$

$\wedge$

ε

.

δ

S

q

/

δ

φ

$^2]$

φ

$$= 0$$

This is the **Entropic Klein-Gordon Equation**, which reduces to the standard Klein-Gordon equation $\square(\square + m^2)\varphi = 0$ when $\Lambda_\varepsilon \rightarrow 0$ (no cosmological coupling) and $q \rightarrow 1$ (extensive entropy). The Tsallis term introduces a nonlinear self-interaction for matter fields driven by the entropy of the field itself — a self-interaction that is responsible for the Higgs mechanism in the ToE interpretation (see Section 9.3).

4.4 Limiting Cases of the Obidi Action

The Obidi Action contains all known physics as limiting cases:

Limit	Conditions	Result
Einstein-Hilbert action	$\beta \rightarrow 0, \alpha \rightarrow 0, q \rightarrow 1$	$S_{EH} = \int R\sqrt{-g} \, d^4x / (16\pi G) + \Sigma \text{ matter terms}$
Standard Model action	$q \rightarrow 1, \text{curvature} \rightarrow 0$	$S_{SM} = \int [-\frac{1}{4}F^2 + \bar{\psi}(i\gamma\partial - m)\psi + D\varphi ^2 - V(\varphi)] \, d^4x$
Quantum information theory	$G_\varepsilon \rightarrow 0, \text{curvature} \rightarrow 0$	$S_{QI} = \beta \int \text{Tr}(\rho \ln \rho) \, dV$ (von Neumann entropy action)
Verlinde entropic gravity	Weak field, non-relativistic, $q \rightarrow 1$	$F = T(\Delta S / \Delta x)$ on holographic screen
Tsallis statistical mechanics	$G_\varepsilon \rightarrow 0, \alpha \rightarrow 0, \text{flat } \mathbb{N}$	Standard Tsallis non-extensive statistical mechanics

The fact that all known fundamental action principles are limiting cases of the Obidi Action constitutes the primary unification claim of the Theory of Entropicity. It is not merely that ToE is "consistent" with known physics; rather, known physics is *derived* from ToE through specific limiting procedures, in precisely the same way that Newtonian mechanics is derived from general relativity in the limit of weak fields and low velocities.

5. The Master Entropic Equation

5.1 Statement of the Master Entropic Equation

Varying the Obidi Action (4.1) with respect to the entropic metric $g^{(\varepsilon)}_{\mu\nu}$ and setting the variation to zero, we obtain the **Master Entropic Equation**:

(5.1)

G

(ε)

$\mu\nu$

+

Λ

ε

g

(ε)

$\mu\nu$

+

Φ

$\mu\nu$

(S

q

) = (8

π

G

ε

$/c^4) \cdot [T$

(matter)

$\mu\nu$

+ T

(entropy)

$\mu\nu$

+ T

(information)

$\mu\nu$

]

The individual tensors in the Master Entropic Equation are defined as follows:

Left-hand side (geometric/informational):

(5.2)

G

(ε)

$\mu\nu$

= R

(ε)

$$\mu\nu$$

$$-$$

$$\tfrac{1}{2}\,g$$

$$(\varepsilon)$$

$$\mu\nu$$

$$\mathfrak{R}$$

$$\varepsilon$$

is the **Entropic Einstein tensor**, built from the Ricci tensor $R^{(\varepsilon)}_{\mu\nu}$ and Ricci scalar $\mathfrak{R}_\varepsilon$ of the Fisher-Rao manifold $\mathbb{N}$. It encodes the information-geometric curvature of physical spacetime.

$$(5.3)$$

$$\Phi$$

$$\mu\nu$$

$$(S$$

$$\mathfrak{q}$$

$$)= (\mathfrak{q}$$

$$-$$

$$1)[$$

$$\partial$$

$$\mu$$

$$\partial$$

ν

S

q

$-$

g

(ε)

$\mu\nu$

$\square S$

q

$]$

is the **Tsallis correction tensor**, encoding the effects of non-extensivity on the geometry of spacetime. When $q = 1$ (extensive entropy), $\Phi_{\mu\nu} = 0$ exactly, and the left-hand side reduces to the Einstein tensor plus cosmological term of standard GR.

Right-hand side (physical sources):

(5.4)

T

(entropy)

$\mu\nu$

$=$

$-$

$T[$

∂ μ S ∂ ν S $-$ $\frac{1}{2}g$ (ε) $\mu\nu$ $($ ∂ $S)^2]$

is the **entropy stress tensor**, representing the contribution of entropy gradients to the physical stress-energy. It is the entropic analogue of the energy-momentum tensor of a scalar field, with S playing the role of the field. This term has no analogue in standard GR.

 (5.5) T (information) $\mu\nu$ $= (\hbar c/4$

π $) \cdot g$ (F) $\mu\nu$

is the **Fisher information tensor**, representing the contribution of information density to the physical stress-energy. This term is proportional to $\hbar$ and vanishes in the classical limit — it encodes the intrinsically quantum-mechanical informational content of spacetime.

5.2 Physical Interpretation

The Master Entropic Equation (5.1) has a deep physical interpretation that differs fundamentally from that of the Einstein field equations. In Einstein's equations, the left-hand side (geometry) is determined by the right-hand side (matter-energy). In the Master Entropic Equation:

The **left-hand side** represents the *geometry of information*: it encodes how entropy — distributed across the entropic manifold — curves the information space. The Tsallis correction tensor $\Phi_{\mu\nu}$ adds non-extensive corrections that become significant wherever entropy is strongly non-extensive: near black holes, in the early universe, and in systems with strong long-range correlations.

The **right-hand side** contains three qualitatively distinct sources of spacetime curvature:

6. **Matter:** the familiar energy-momentum tensor of standard GR.
7. **Entropy:** the entropy-stress tensor, representing the direct gravitational effect of entropy gradients. This term explains dark matter without new particles: the entropy gradients in galactic halos — arising from the non-extensive statistics of dark halos — produce a $T^{(\text{entropy})}_{\mu\nu}$ that mimics the gravitational effect of missing mass.

8. **Information:** the Fisher information tensor, representing the quantum-mechanical informational content of spacetime. This term explains the vacuum energy / dark energy: the Fisher information of the de Sitter vacuum is non-zero and provides the cosmological constant Λ_ε with a natural value proportional to $\hbar/V_{\text{Hubble}}$.

5.3 Conservation Laws from the Master Entropic Equation

The diffeomorphism invariance of the Obidi Action implies, via Noether's theorem, a fundamental conservation law for the Master Entropic Equation. The entropic Bianchi identity on $\mathbb{N}$ states:

(5.6)

∇

μ

G

$(\varepsilon)\mu\nu$

$= 0$

Applied to the Master Entropic Equation, this yields:

(5.7)

∇

μ

$[T$

$(\text{matter})\mu\nu$

$+ T$

(entropy) $\mu\nu$

+ T

(information) $\mu\nu$

] =

-

($c^4/8$

π

G

ε

) ·

∇

μ

[

$\wedge$

ε

g

(ε) $\mu\nu$

+

Φ

$\mu\nu$

]

This is the **total conservation law of ToE**: the total stress-energy-entropy-information is covariantly conserved (in the generalized sense that includes the time-varying Λ_ε and Φ corrections). The second law of thermodynamics — the non-decrease of total entropy — is a consequence of this identity: it is the timelike component of (5.7), stating that the total entropy current $\nabla_\mu S^\mu \geq 0$ in any physical process. The thermodynamic arrow of time is therefore not an additional postulate in ToE — it is a geometric identity on the entropic manifold $\mathbb{N}$, following from the same diffeomorphism invariance that gives rise to the Bianchi identity in GR.

6. The Entropic Einstein Equation

6.1 Derivation from the Master Equation

To make contact with standard general relativity, we take two successive limits of the Master Entropic Equation. First, the **extensive limit** $q \rightarrow 1$, in which the Tsallis non-extensive corrections vanish: $\Phi_{\mu\nu} \rightarrow 0$. Second, the **classical limit** $\hbar \rightarrow 0$, in which the Fisher information tensor vanishes: $T^{(\text{information})}_{\mu\nu} \rightarrow 0$. In these limits, the Master Entropic Equation reduces to:

(6.1)

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8$$

π $G/c^4) \cdot [T$ (matter) $\mu\nu$ $+ T$ $(\text{entropy correction})$ $\mu\nu$ $]$

This is the **Entropic Einstein Equation**. The entropy-correction tensor retains a non-zero classical contribution:

 (6.2) T $(\text{entropy correction})$ $\mu\nu$ $= (c^4/4G) \cdot [$ ∂ μ ∂ ν S $-$

$$g_{\mu\nu} \square S] - (k_B T/4G) \cdot R_{\mu\nu}$$

The first term represents the direct contribution of entropy gradients to the spacetime curvature: it is the classical entropic stress. The second term represents the backreaction of curvature on entropy through the temperature T — in hotter regions of spacetime, the curvature-entropy coupling is stronger. Both terms vanish when S is at its maximum (equilibrium), or when S is uniform across spacetime.

6.2 Relationship to Standard General Relativity

The recovery of standard GR from the Entropic Einstein Equation occurs in the **entropic equilibrium limit**: when the entropy of the physical system is at its maximum given the constraints ($T^{(\text{entropy correction})}_{\mu\nu} \rightarrow 0$), the Entropic Einstein Equation reduces exactly to:

$$(6.3) \quad G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$$

g

$\mu\nu$

$= (8$

π

$G/c^4) T$

(matter)

$\mu\nu$

which is the standard Einstein equation with cosmological constant. Standard GR is thus an exact result of ToE in the limit of maximum entropy — a limit corresponding to spacetimes in thermodynamic equilibrium. This provides a precise answer to why GR works so well: the spacetimes we observe are very close to entropic equilibrium at the scales probed by classical gravitational experiments.

Away from equilibrium, the Entropic Einstein Equation predicts corrections to GR. For the Schwarzschild black hole, the entropic correction modifies the metric as:

(6.4)

$ds^2 =$

$-$

$(1$

$-$

$2GM/rc^2 +$

ϵ

S

·S

BH

$/r^2)c^2dt^2 + (1$

–

$2GM/rc^2 +$

ε

S

·S

BH

$/r^2)^{-1}dr^2 + r^2d$

Ω

2

where $\varepsilon_s = k_B/(4\pi c^2)$ is a small entropic correction coefficient and S_{BH} is the Bekenstein-Hawking entropy. This correction is sub-leading in r and vanishes at large distances, ensuring consistency with solar system tests of GR. Near the horizon, the entropic correction becomes significant and modifies the near-horizon geometry in a way that resolves the information paradox: information is not destroyed but is encoded in the entropic correction to the metric, which imprints itself on the outgoing Hawking radiation.

The **Bekenstein-Hawking entropy formula** $S_{\text{BH}} = Ak_{\text{B}}c^3/(4G\hbar)$ emerges naturally from ToE as the value of the von Neumann entropy of the entropic metric at the horizon — the entropy encoded in the Fisher information of the near-horizon probability distributions. This provides a first-principles derivation of the Bekenstein-Hawking formula from information geometry, without appeal to string theory, loop quantum gravity, or other auxiliary structures [9, 10].

6.3 Verlinde's Entropic Gravity as a Special Case

Verlinde's entropic gravity program [15, 16] derives the gravitational force as $F = T(\Delta S/\Delta x)$, where the entropy S is associated with a holographic screen surrounding a mass M . In the weak-field, non-relativistic limit of the Entropic Einstein Equation (6.1), taking M small, speeds $v \ll c$, and neglecting entropy corrections in the $T^{(\text{entropy correction})}$ term except at leading order, we obtain:

(6.5)

∇

2

Φ

$= 4$

π

G

ρ

$+ T \cdot$

∇

$^2 S$

where Φ is the Newtonian gravitational potential and ρ is the matter density. For a holographic screen at radius R surrounding mass M , the entropy is $S_{\text{screen}} = 2\pi k_B M c \cdot R / \hbar$, giving $\nabla S = 2\pi k_B M c / (\hbar R^2) \cdot \hat{R}$, and the entropic force $F = T \cdot \nabla S$ reproduces Newton's law $F = GMm/r^2$ exactly, confirming Verlinde's result as the weak-field limit of the Entropic Einstein Equation.

The Theory of Entropicity extends Verlinde's program in three critical ways. First, it provides a fully covariant formulation valid in strong-field regimes, including near black holes and in the early universe. Second, it incorporates quantum corrections

$$(7.1) \quad \frac{dS}{d\tau} \leq v(\max) \varepsilon = (k$$

$$= (k$$

B

$c/\hbar) \cdot E$

system

$= (2$

π

k

B

$/\hbar) \cdot T \cdot E$

system

Furthermore, no physical process can force the system to evolve faster than the entropic gradient allows without violating the Master Entropic Equation. The bound is saturated if and only if the system is in a pure state (zero von Neumann entropy) evolving unitarily between orthogonal states.

The No-Rush Theorem constitutes the entropic analogue of the quantum speed limit. In quantum mechanics, the Margolus-Levitin theorem [22] states that the minimum time $\tau_{\min}$ for a quantum system to evolve from one state to an orthogonal state is $\tau_{\min} = \pi\hbar/(2E)$, where E is the mean energy above the ground state. The No-Rush Theorem is the entropic generalization and deepening of this result: it states that not just quantum transitions but all physical evolution — including classical, gravitational, and cosmological evolution — is bounded in its rate of entropic change by the energy of the system.

7.2 Proof Sketch

Proof. Consider a physical trajectory $\gamma(\tau)$ on the entropic manifold $\mathbb{N}$, parametrized by proper time τ . The rate of entropic change along γ is:

$$\begin{aligned}
 (7.2) \quad & \frac{dS}{d\tau} \\
 &= \left(\frac{\partial S}{\partial q_i} \right) \cdot \left(\frac{dq_i}{d\tau} \right) \\
 &= \nabla S \cdot \frac{d\gamma}{d\tau}
 \end{aligned}$$

Applying the Cauchy-Schwarz inequality to the Fisher-Rao inner product:

(7.3)

$|dS/d$

τ

$|^2 = |(\$

∇

$S) \cdot (dq/d$

τ

$)|^2$

$\leq$

g

(F)

ij

$($

∇

$S)$

i

$($

∇

$S)$

j

$\cdot g$

(F)

kl

$(dq$

k

$/d$

τ

$)(dq$

l

$/d$

τ

$)$

The first factor, $g^{(F)}_{ij}(\nabla S)^i(\nabla S)^j$, is the squared Fisher norm of the entropy gradient — it measures the maximum statistical distinguishability rate. By the Cramér-Rao bound in information geometry, this is bounded above by the quantum Fisher information F_Q of the state $\rho(\tau)$:

(7.4)

g

(F)

ij

$($

∇

$S)$

i

$$\begin{aligned} & \left(\nabla^2 S \right)_{jj} \\ & \leq 4 \cdot \text{Var}(\hat{H})/\hbar^2 = 4 \\ & \Delta E^2/\hbar^2 \end{aligned}$$

The second factor, $g^{(F)}_{kl}(dq^k/d\tau)(dq^l/d\tau)$, is the squared speed of the trajectory in the Fisher metric — bounded by the Bures metric geodesic speed. The geodesic length in the Bures metric satisfies:

$$\begin{aligned} & (7.5) \\ & \left| \frac{d}{d\tau} \right|_{\text{Bures}} \\ & = \int \sqrt{g^{(F)}_{kl}} \end{aligned}$$

kl

$(dq$

k

$/d$

τ

$)(dq$

l

$/d$

τ

$)] d$

τ

$\leq$

2

$\int$

Δ

$E/\hbar d$

τ

Combining (7.3), (7.4), and (7.5):

(7.6)

$|dS/d$

τ

$$\begin{aligned}
 &| \\
 &\leq \\
 &(2 \\
 &\Delta \\
 &E/\hbar) \cdot (2 \\
 &\Delta \\
 &E/\hbar)^{1/2} \cdot C \\
 &\leq \\
 &(2 \\
 &\pi \\
 &k \\
 &B \\
 &T/\hbar) \cdot E
 \end{aligned}$$

where the last inequality uses the thermal equilibrium relation $\Delta E = k_B T$ for a system at temperature T and the Margolus-Levitin bound $E \geq \hbar\pi/(2\tau_{\min})$ [22]. This completes the proof. $\square$

7.3 Consequences of the No-Rush Theorem

Time emergence. The No-Rush Theorem implies that coordinate time t is the integral of entropic flow: $t = \int (dS/d\tau)^{-1} dS$. "Time," in ToE, does not exist as a fundamental quantity — it is the accumulation of entropic changes. A system in exact thermodynamic equilibrium ($dS/d\tau = 0$) experiences no time — a result that resonates with Barbour's "End of Time" program [30] and the timelessness of the Wheeler-DeWitt equation.

Quantum speed limit. The Margolus-Levitin theorem [22] and the Mandelstam-Tamm bound $\tau_{\min} \geq \hbar/2\Delta E$ are special cases of the No-Rush Theorem for pure quantum states. The No-Rush Theorem extends these results to mixed states, curved spacetimes, and systems with non-extensive entropy.

Cosmic expansion. The Hubble parameter $H(t)$ is, in ToE, the rate of entropic expansion of the observable universe. The No-Rush Theorem bounds H from above: $H \leq (2\pi k_B/\hbar) \cdot T_{\text{CMB}} \cdot E_{\text{Hubble}}$, where $T_{\text{CMB}} \approx 2.7$ K is the CMB temperature and E_{Hubble} is the total energy within the Hubble horizon. The second law of thermodynamics bounds H from below: $H \geq 0$ (entropy never decreases). The observed cosmic expansion rate is sandwiched between these entropic bounds, providing a natural explanation of why the universe expands at its observed rate without fine-tuning.

Consciousness and computation. Any computational process — including neural computation — that involves a change of entropic state is subject to the No-Rush bound. The minimum time required for a computational operation that changes entropy by ΔS , carried out by a system at temperature T with total energy E , is:

(7.7)

τ

min

(computation) = $\hbar$

Δ

$S / (2$

π

k

B

$TE)$

This provides a fundamental physical lower bound on the speed of thought — and a universal limitation on artificial intelligence, biological cognition, and cosmological computation alike. Thought, in the Theory of Entropicity, has an entropic speed limit.

8. The Computational Universe Interpretation

8.1 Wheeler's "It from Bit"

Wheeler's 1990 essay "Information, Physics, Quantum: The Search for Links" [1] proposed that every physical entity — every particle, every field of force, even spacetime itself — derives its function, existence, and meaning from the answers to binary yes-or-no questions. Wheeler encapsulated this vision in the phrase "It from Bit": physical reality (It) emerges from information (Bit). This proposal was visionary but programmatic: Wheeler did not specify the mathematical mechanism by which binary information could generate the specific structures of relativistic field theory and quantum mechanics.

The Theory of Entropicity provides this mechanism. In ToE, bits are the $q \rightarrow 1$ (extensive) limit of **entropic qubits** — units of information measured in the natural units of the entropic manifold. The universe is not merely informational in Wheeler's sense; it is **entropic-computational**: every physical interaction is an entropic computation that updates the state of the entropic manifold $\mathbb{N}$ in accordance with the Master Entropic Equation. Physical laws are not constraints imposed on a pre-existing spacetime; they are the rules of this entropic computation — the geodesic equations, conservation laws, and field equations of $\mathbb{N}$.

Wheeler's **Participatory Universe** — in which observers play an active, constitutive role in creating physical reality through acts of observation — is realized in ToE as the statement that observation is an *entropic collapse*: the act of observation selects a definite microstate from the entropic superposition encoded in the probability distribution $p(x|q)$, reducing the entropy of the observed subsystem and increasing

the entropy of the observer-environment. The universe is participatory because every observation is a computation, and every computation modifies $\mathbb{N}$.

8.2 Wolfram's Ruliad and Entropic Computation

Wolfram's 2020 Wolfram Physics Project [28] identifies the universe with the **Ruliad** — the maximal entangled limit of all possible computational rules — and proposes that physical laws are the rules perceived by observers embedded within this structure. The Ruliad is defined as the infinite limit of all possible hypergraph rewriting systems, and different physical laws correspond to different "slices" of the Ruliad accessible to particular types of observers.

In ToE, we provide a natural mathematical identification: **the Ruliad is the entropic manifold $\mathbb{N}$** . The correspondence is precise:

- Different "computational rules" in Wolfram's sense $\leftrightarrow$ Different regions of $\mathbb{N}$ with different entropic curvatures $\mathfrak{R}_\varepsilon$.
- Wolfram's "observer threads" $\leftrightarrow$ Entropic sub-manifolds of $\mathbb{N}$ that sample consistent, decoherent slices of the entropic superposition.
- Wolfram's computational irreducibility $\leftrightarrow$ Regions of maximal entropic curvature on $\mathbb{N}$ where the geodesic equations cannot be integrated in closed form — the entropic manifold is itself computationally irreducible in these regions.
- Wolfram's causal graph $\leftrightarrow$ The causal structure of $\mathbb{N}$, determined by the entropic connection (3.12).

This identification provides a significant advantage over Wolfram's program: it equips the Ruliad with a specific, well-defined geometric structure (the Fisher-Rao metric) and a specific dynamics (the Master Entropic Equation), rather than leaving these as emergent from an unspecified rewriting system. ToE is, in this sense, a completion of Wolfram's program.

8.3 The Entropic Feedback Loop

A central structural feature of the Theory of Entropicity is the **Entropic Feedback Loop**: the dynamical process by which the entropy production of a physical system

modifies the geometry of $\mathbb{N}$, which in turn modifies the system's dynamics, generating further entropy production.

Mathematically, the Entropic Feedback Loop is described by the autonomous dynamical system on the space of entropic metrics:

$$\begin{aligned} (8.1) \quad & \frac{dg}{dt} = \\ & -\kappa \cdot [T^{(entropy)}] \\ & - \frac{1}{3}g^{(\epsilon)}_{\mu\nu} T^{(entropy)}_{\mu\nu} \end{aligned}$$

where $\kappa = 8\pi G_{\epsilon}/c^4$ is the entropic Einstein constant and $T^{(entropy)} = g^{\mu\nu}T^{(entropy)}_{\mu\nu}$ is the trace of the entropy stress tensor. Equation (8.1) is an autonomous dynamical system on the space of Riemannian metrics $\mathcal{M} = \{g^{(\epsilon)} : \text{solutions of Master Entropic}$

Equation}, and its fixed points correspond to the stable physical configurations of spacetime:

Fixed Point	Condition	Physical Meaning
Minkowski spacetime	$T^{(\text{entropy})}_{\mu\nu} = 0, \mathfrak{R}_\epsilon = 0$	Vacuum: zero entropy gradient, flat entropic manifold
Schwarzschild black hole	$S = S_{\text{max}}, T^{(\text{entropy})}_{\mu\nu} = T_{\text{Hawking}} T_{\mu\nu}$	Maximum entropy state; attractor of gravitational collapse
de Sitter spacetime	$\Lambda_\epsilon > 0, \mathfrak{R}_\epsilon = \text{const}$	Entropic equilibrium with cosmological constant; late-universe attractor

The Big Bang as feedback instability. In ToE, the origin of the universe is understood as a spontaneous symmetry-breaking instability of the entropic vacuum — the Minkowski fixed point of the entropic feedback loop. Quantum fluctuations of the entropic metric near the Minkowski fixed point are generically unstable at trans-Planckian scales, where the Fisher information term $T^{(\text{information})}_{\mu\nu}$ dominates. Above the Planck energy $E_{\text{Pl}} = \sqrt{(\hbar c^5/G)} \approx 1.2 \times 10^{19} \text{ GeV}$, the entropic feedback loop undergoes a phase transition: the Minkowski fixed point becomes unstable, and the system cascades rapidly toward the high-curvature (high-entropy-production) regime. This cascade is the Big Bang — an entropic explosion, not an energy explosion.

Attractor analysis. The long-time behavior of the entropic feedback loop is governed by the attractor structure of the dynamical system (8.1). Using Lyapunov analysis, one can show that:

- The Minkowski fixed point is an unstable saddle (quantum fluctuations drive it away).
- The de Sitter fixed point is a stable attractor (any spacetime with $\Lambda_\epsilon > 0$ flows toward de Sitter at late times).
- Black holes are local attractors in the matter sector (gravitational collapse is an entropic downhill flow toward S_{max}).

- The cosmic evolution from Big Bang to heat death is a trajectory in $\mathcal{M}$ from the unstable Minkowski point (near-infinite curvature), through a saddle at the de Sitter point (dark-energy-dominated expansion), toward the stable attractor of maximum entropy (flat, zero-temperature vacuum).

8.4 The Universe as Entropic Computer

Every physical interaction in ToE is an **entropic computation**: it reads the current state of the entropic manifold $\mathbb{N}$, applies the Master Entropic Equation (as a "rule"), and updates $\mathbb{N}$ to a new state. This is not a metaphor — it is a precise mathematical statement. The entropic feedback loop (8.1) is a dynamical system, and dynamical systems are, in the computational sense, universal computers whenever their dynamics are computationally irreducible.

Landauer's principle in ToE. Landauer (1961) [21] showed that the erasure of one bit of information in a physical memory device requires a minimum energy dissipation of $k_B T \ln 2$ joules at temperature T . In ToE, this result is a theorem: it follows directly from the entropic Bianchi identity (5.6), which requires that any decrease in the information content of a subsystem (erasure of information) must be compensated by an equal increase in the entropy of the environment. The minimum heat dissipated is exactly $k_B T \ln 2$, confirming Landauer's principle as a necessary consequence of the Master Entropic Equation. This was experimentally confirmed by Bérut et al. (2012) [24].

The Bekenstein bound as computational complexity. The Bekenstein bound [9] states that the maximum entropy of a physical system of radius R and energy E is $S \leq 2\pi k_B R E / (\hbar c)$. In ToE, this bound is the *computational complexity bound* on the entropic manifold: it limits the number of distinguishable states of a physical system, and hence the number of distinct computations it can perform per unit volume. The Bekenstein bound is the information-geometric isoperimetric inequality of $\mathbb{N}$: the maximum entropy is proportional to the Fisher-Rao volume of the boundary of the system's entropic sub-manifold.

The holographic principle. The Bekenstein-Hawking area law and the holographic principle [25, 27] state that the information content of a region of spacetime is

proportional to the area of its boundary, not its volume. In ToE, this is the isoperimetric inequality of the entropic manifold: the maximum entropy (= maximum Fisher-Rao volume) of a region of $\mathbb{N}$ is bounded by the Fisher-Rao area of its boundary. The holographic principle is thus the entropic manifold's natural geometric statement about the relationship between entropy and surface area — a statement that follows from the structure of the Fisher-Rao metric without additional assumptions.

9. Connections to Quantum Field Theory

9.1 Quantum Fields as Entropic Fluctuations

In standard quantum field theory, quantum fields $\varphi(x)$ are operator-valued distributions on a fixed spacetime background, with dynamics governed by a Lagrangian that specifies interactions. In the Theory of Entropicity, quantum fields are not independently existing entities — they are **fluctuations of the entropic manifold $\mathbb{N}$ around its ground state**.

More precisely, let $\mathbb{N}_0$ denote the ground-state (vacuum) configuration of the entropic manifold — the stable, lowest-entropy-production fixed point of the entropic feedback loop. Fluctuations $\delta g^{(e)}_{\mu\nu}$ around $\mathbb{N}_0$ correspond to quantum fields:

(9.1)

φ

(x)

$\leftrightarrow$

δ

g

(ε)

$\mu\nu$

(x) on

$\mathbb{N}$

near

$\mathbb{N}$

0

The path integral of quantum field theory:

(9.2)

$Z =$

$\int$

D

φ

$\exp(iS$

matter

$[$

φ

$]/\hbar)$

is identified in ToE with the **Boltzmann weight of entropic configurations** after a Wick rotation $t \rightarrow -i\tau$ (Euclidean signature):

(9.3)

$Z =$

$\int$

Dg

(ε)

$\exp($

$-$

S

Obidi

$[g$

(ε)

$] / k$

B

$)$

This identification reveals the deep physical meaning of the Wick rotation: it is not merely a mathematical trick for regularizing divergences, but a *physical rotation from Lorentzian (real-time) to Euclidean (thermal) entropic manifold*. The imaginary time $\tau = it$ of quantum field theory is the thermal time of the entropic manifold — the time coordinate appropriate for describing thermal, entropic evolution rather than coherent quantum evolution.

9.2 Renormalization as Entropic Coarse-Graining

One of the most powerful tools of quantum field theory is the **renormalization group (RG)**, which describes how the effective parameters of a theory change as we

vary the energy scale at which we probe the system. In ToE, the RG flow has a natural and beautiful interpretation: it is **entropic coarse-graining on $\mathbb{N}$** .

When we "integrate out" high-energy (short-wavelength) modes of a quantum field to obtain an effective theory at lower energies, we are, in ToE, averaging over small-scale fluctuations of the entropic manifold. This averaging increases the effective non-extensivity parameter q , since short-scale correlations are lost in the coarse-graining. Specifically:

(9.4)

$dq/d(\ln$

μ

$) =$

β

ε

(q, g)

$<$

0

where β_ε is the **entropic beta function** governing the running of q with the energy scale μ . At high energies (UV), q is large: entropy is strongly non-extensive, reflecting the strong correlations of the quantum vacuum at short scales. At low energies (IR), $q \rightarrow 1$: entropy becomes extensive, reflecting the thermodynamically independent degrees of freedom of the classical world.

The UV divergences of quantum field theory — the infinite renormalization of coupling constants that requires regularization and renormalization — correspond in ToE to the $q \rightarrow \infty$ limit (extreme non-extensivity at the Planck scale). The Tsallis cutoff at the Planck scale naturally regularizes these divergences: for $q > q_{\max} = c^4/(G\hbar\Lambda_{\text{Pl}})$,

the Tsallis entropy functional is itself cut off by the entropic structure of the manifold, providing a natural, physical ultraviolet regularization without the need for dimensional regularization or Pauli-Villars regulators.

The running coupling constant $g(\mu)$ of a gauge theory is identified in ToE with the entropic coupling between the gauge field and the Fisher-Rao metric:

$$(9.5) \\ g(\mu) = g_0 \cdot \exp\left[-\frac{\beta}{\epsilon} \cdot S_q(\mu)\right]$$

where $S_q(\mu)$ is the Tsallis entropy of the modes with energy scale μ . This formula reproduces the standard asymptotic freedom of QCD (g decreases at high μ , since S_q increases with μ for $q > 1$) and the infrared slavery (g increases at low μ) as special cases.

9.3 The Standard Model from Entropic Gauge Theory

In ToE, the gauge symmetries of the Standard Model — the $SU(3) \times SU(2) \times U(1)$ gauge group — are symmetries of the entropic manifold $\mathbb{N}$ under reparametrization of the probability distributions. Specifically, they are the automorphisms of the Tsallis entropy functional S_q that leave S_q invariant while permuting the probability distributions p_i among themselves.

The correspondence is as follows. The **U(1) electromagnetic gauge symmetry** corresponds to the phase symmetry of complex probability amplitudes $\psi = \sqrt{p} \cdot e^{i\theta}$ on $\mathbb{N}$: the Fisher-Rao metric is invariant under $\theta \rightarrow \theta + \text{const}$, just as electromagnetism is invariant under $\varphi \rightarrow \varphi \cdot e^{i\theta}$. The **SU(2) weak gauge symmetry** corresponds to the rotation symmetry of doublets of entropic states — pairs of probability distributions related by a Tsallis-symmetric operation. The **SU(3) strong gauge symmetry** corresponds to the permutation symmetry of triplets of entropic states (the entropic analogue of quark color).

The **Higgs mechanism** in ToE is the spontaneous breaking of Tsallis entropy extensivity. In the high-temperature (early universe) limit, all entropy is extensive ($q = 1$) and all gauge symmetries are unbroken — all particles are massless. As the universe cools, the Tsallis parameter q departs from 1 (entropy becomes non-extensive), the Obidi Action develops a non-trivial minimum at $q \neq 1$, and the gauge symmetry is spontaneously broken — particles acquire mass through the Higgs mechanism, which is, in ToE, the mechanism of entropic symmetry breaking.

10. Comparative Analysis

10.1 ToE vs. General Relativity

Feature	General Relativity	Theory of Entropicity
Fundamental entity	Spacetime metric $g_{\mu\nu}$	Entropic manifold $\mathbb{N}$ with Fisher-Rao metric $g^{(\epsilon)}_{\mu\nu}$

Feature	General Relativity	Theory of Entropicity
Source of curvature	Energy-momentum tensor $T_{\mu\nu}$	Matter + entropy stress + Fisher information tensors
Field equation	$G_{\mu\nu} = 8\pi G T_{\mu\nu}/c^4$	Master Entropic Equation (5.1) — richer structure
Geodesics	Minimize proper time	Minimize entropic distance on $\mathbb{N}$
Singularities	At $r = 0$ (geodesic incompleteness)	Absent: entropic manifold is complete (Fisher metric regular)
Information paradox	Unresolved: information lost in black holes	Resolved: information encoded in entropic correction to metric
Quantum gravity	Not provided	Via spectral algebra of $\hat{E}$ on $\mathbb{N}$; natural quantization
Dark matter	Requires new particle species	$T^{(\text{information})}_{\mu\nu}$ explains rotation curves without new particles
Dark energy	Requires fine-tuned Λ	$\Lambda_\epsilon = S_q(\text{vacuum})/V$ — dynamically determined

10.2 ToE vs. Quantum Field Theory

Feature	Quantum Field Theory	Theory of Entropicity
Nature of fields	Operator-valued distributions on fixed spacetime	Fluctuations of the entropic manifold $\mathbb{N}$
Renormalization	Ad hoc: required by UV divergences	Natural: entropic coarse-graining with Tsallis cutoff
Measurement problem	Unresolved: collapse is postulated	Measurement = entropic collapse to definite microstate on $\mathbb{N}$
Free parameters	19 free parameters in Standard Model	Fixed by q, α, β of Obidi Action (3 fundamental parameters)
Hierarchy problem	Unresolved: Higgs mass sensitive to UV physics	Natural: Tsallis cutoff suppresses UV sensitivity
CPT symmetry	Assumed as fundamental axiom	Derived: symmetry of Fisher-Rao metric under time-reversal
Unification with gravity	Not achieved	Unified via Obidi Action on

Feature	Quantum Field Theory	Theory of Entropicity
		single entropic manifold

10.3 ToE vs. Verlinde's Entropic Gravity

Feature	Verlinde Entropic Gravity	Theory of Entropicity
Gravity mechanism	Entropic force on holographic screen	Geodesic on entropic manifold $\mathbb{N}$
Entropy measure	Bekenstein-Hawking (area law)	Tsallis q-entropy (subsumes B-H as $q \rightarrow 1$ limit)
Covariance	Non-covariant in original form	Fully covariant via Obidi Action
Quantum extension	Not provided	Spectral algebra of $\hat{E}$ provides natural quantization
Strong-field regime	Not applicable	Fully valid (Entropic Einstein Equation)
Matter fields	External to the framework	Naturally included via $\mathcal{L}_{\text{matter}}$ in Obidi Action

10.4 ToE vs. Loop Quantum Gravity and String Theory

Feature	Loop Quantum Gravity	String Theory	Theory of Entropicity
Spacetime at Planck scale	Discrete (spin networks)	Replaced by string geometry	Continuous $\mathbb{N}$ with discrete $\hat{E}$ spectrum
Extra dimensions	None	6-7 compactified extra dimensions	Directions on $\mathbb{N}$ (probability space)
Status of entropy	Derived quantity	Derived (via holography)	Ontological primitive
Prediction power	Limited	Landscape problem: 10^{500} vacua	Specific predictions from Obidi Action
Matter unification	Not unified	Partially unified	All forces from $SU(3) \times SU(2) \times U(1)$ of $\mathbb{N}$
Background independence	Yes	No (perturbative)	Yes: Obidi Action is diffeomorphism-invariant on $\mathbb{N}$

11. Predictions and Experimental Signatures

11.1 Observational Predictions

Modified dispersion relation. The Fisher information tensor $T^{(\text{information})}_{\mu\nu}$ in the Master Entropic Equation modifies the dispersion relation of photons traveling through the entropic manifold. The predicted modification is:

(11.1)

$$E^2 = p^2 c^2 + m^2 c^4 +$$

η

$\cdot (k$

B

T

CMB

$/E$

Planck

$) \cdot E^3/c^2$

where $\eta = O(1)$ is a dimensionless parameter fixed by α in the Obidi Action and $E_{\text{Planck}} = \sqrt{(\hbar c^5/G)} \approx 1.22 \times 10^{19} \text{ GeV}$. This energy-dependent photon speed produces a time delay $\Delta t \approx \eta \Delta E \cdot D / (2 E_{\text{Planck}} c)$ between high- and low-energy photons from a gamma-ray burst at luminosity distance D . For $D \sim 10 \text{ Gpc}$ and $\Delta E \sim 100 \text{ GeV}$, the predicted delay is $\Delta t \sim 1\text{-}10 \text{ ms}$ — within the detection capability of the Fermi Gamma-Ray Space Telescope (FGST). Current FGST data already constrains $\eta \leq$

$O(1)$ for the linear term, and next-generation telescopes (CTA) will probe η to sub-leading order.

Entropic dark energy equation of state. The identification $\Lambda_e = S_q(\text{vacuum})/V$ in the Obidi Action predicts a time-varying dark energy equation of state:

(11.2)

$$w(z) = p$$

DE

/

ρ

DE

=

–

$1 + (q(z)$

–

$1) \cdot f(z)$

where z is the cosmological redshift and $f(z)$ is a function of the Tsallis parameter's redshift evolution. For q constant (extensive entropy), $w = -1$ exactly (cosmological constant). For q evolving as the universe expands (entropic coarse-graining via RG flow), $w(z) \neq -1$ — a signature testable by the EUCLID satellite (launched 2023), which aims to measure $w(z)$ to 1% precision over $0 < z < 2$, and by the Rubin Observatory LSST.

Gravitational wave echoes. The entropic correction to the Schwarzschild metric (6.4) produces a modified near-horizon reflectivity for gravitational waves. After a binary black hole merger, the post-ringdown gravitational wave signal should

contain sub-leading **entropic echoes** — reflections of the gravitational waves from the entropic-corrected near-horizon geometry — at a time delay:

(11.3)

$$\Delta t_{\text{echo}} \approx \frac{4GM}{c^3} \cdot \left| \ln \left(\frac{\epsilon}{S_{\text{BH}}} \right) \right| + O(G^2 M^2 / M^2 c^4)$$

For stellar-mass black holes ($M \sim 30 \text{ } M_{\odot}$), $\Delta t_{\text{echo}} \sim 0.1\text{-}1 \text{ s}$ — potentially detectable with LISA or third-generation ground-based detectors (Einstein Telescope, Cosmic Explorer).

11.2 Laboratory Predictions

Tsallis statistics in quantum systems. Non-extensive Tsallis statistics ($q \neq 1$) produce anomalous heat capacity scaling in systems with long-range correlations:

(11.4)

$$C$$

V

$(T) \sim T$

$(3+q-1)$

for q

$\neq$

1 (vs. C

V

$\sim T^3$ for phonons at $q = 1$)

This is measurable in ultracold atomic gases (Bose-Einstein condensates) with tunable interactions, where q can be varied by adjusting the interaction range via Feshbach resonances. Current precision of heat capacity measurements in ultracold systems ($\sim 1\%$) is sufficient to detect $q - 1 \sim 0.01$, well within the predicted ToE range.

Entropic speed limit. The No-Rush Theorem (7.1) predicts that quantum computation cannot exceed the Margolus-Levitin bound $\Delta t \geq \hbar/(2\Delta E)$. Current superconducting quantum processors operating at $T \sim 15$ mK with qubit energies $E \sim k_B T/2$ operate at approximately 10–20% of the Margolus-Levitin bound, consistent with the No-Rush Theorem. Future quantum computers with higher qubit energies will approach the bound from below, providing a precision test of ToE.

Landauer erasure. The identification of Landauer's principle as a necessary consequence of the entropic Bianchi identity (5.6) predicts that the heat dissipated per erased bit at temperature T is exactly $k_B T \ln 2$ — no corrections from ToE at accessible temperatures, since the Tsallis correction is of order $(q-1)k_B T$, which is negligible at macroscopic temperatures. This is consistent with the high-precision experimental confirmation by Bérut et al. (2012) [24], who measured the Landauer limit to within 5%.

12. The Entropic Feedback Loop: Detailed Analysis

12.1 Mathematical Structure

The Entropic Feedback Loop, introduced in Section 8.3, has a rich mathematical structure closely related to the theory of geometric flows on Riemannian manifolds. The fundamental flow equation (8.1) can be written as a version of the **Entropic Ricci Flow** on $\mathbb{N}$:

(12.1)

$$\frac{\partial}{\partial t} g_{\mu\nu} = -2R_{\mu\nu} + \frac{1}{n} \left(\nabla_\mu \nabla_\nu \mathfrak{R} - \nabla_\mu \nabla_\nu \varepsilon \right)$$

$$\mu\nu$$

This is the normalized Ricci flow equation of Hamilton (1982) [19] and Perelman (2002) [20], now interpreted on the entropic manifold $\mathbb{N}$ rather than a Riemannian manifold of physical space. Perelman's **entropy functional** $W(g, f, \tau)$ has a direct entropic interpretation in ToE:

$$\begin{aligned} (12.2) \\ W(g, f, \\ \tau \\) = \\ \int \\ [\\ \tau \\ (| \\ \nabla \\ f|^2 + R) + f \\ - \\ n] (4 \\ \pi \\ \tau \\)^{\frac{n}{2}} \\ - \end{aligned}$$

$$n/2) e^{(}$$

–

$$f) dV$$

In ToE, $f(x) = -\ln p(x|q)$ is the negative log-probability — precisely the information content of the distribution p at the point x . The functional W is thus the *free energy* of the entropic manifold, and Perelman's monotonicity theorem — W is non-decreasing under Ricci flow — is, in ToE, precisely the second law of thermodynamics: the free energy of the entropic manifold never decreases along the entropic Ricci flow.

The long-time convergence theorem for Ricci flow (Perelman 2002, Hamilton 1982 [19, 20]) states that on compact manifolds of positive curvature, the normalized Ricci flow converges to a constant-curvature (Einstein) metric. In ToE, this means that the long-time behavior of the entropic feedback loop is convergence to an **Einstein metric on $\mathbb{N}$** — a metric in which $R^{(\epsilon)}_{\mu\nu} = \Lambda g_{\mu\nu}$ for some constant Λ . This Einstein metric is precisely the de Sitter metric (for $\Lambda > 0$) — the late-time attractor of cosmic evolution, confirming the cosmological prediction of Section 8.3.

12.2 Cosmological Feedback

The cosmological dynamics of ToE are governed by the entropic generalization of the Friedmann equations. The Hubble parameter $H(t) = \dot{a}/a$ (where $a(t)$ is the cosmological scale factor) is, in ToE, the rate of entropic expansion:

$$(12.3)$$

$$H = (1/3) \cdot d(\ln S$$

total

$$)/dt$$

where S_{total} is the total entropy of the observable universe. This identifies the Hubble expansion rate as an entropic flow rate — the universe expands because its total entropy increases. The Friedmann equations of ToE are:

$$\begin{aligned}
 (12.4) \quad H^2 = & \left(\frac{8\pi G}{3c^2} \right) \cdot \left[\rho_{\text{matter}} + \rho_{\text{entropy}} + \rho_{\text{information}} \right] - k/a^2
 \end{aligned}$$

where $\rho_{\text{entropy}} = T \cdot S/V$ is the entropy energy density and $\rho_{\text{information}} = \hbar c \cdot I_{\text{Fisher}}/(4\pi V)$ is the Fisher information energy density. These additional terms modify the standard cosmological evolution:

Inflation in ToE occurs when the Fisher information density $\rho_{\text{information}}$ dominates near the Planck scale: $I_{\text{Fisher}} \sim V/(\ell_{\text{Pl}})^4$, giving $\rho_{\text{information}} \sim \hbar c/\ell_{\text{Pl}}^4 \sim \rho_{\text{Pl}}$. The resulting exponential expansion $H \sim \text{const}$ satisfies $dS/dt \sim H \cdot S$, giving exponential entropic growth — the entropic analog of inflation.

CMB anisotropies in ToE are frozen entropic fluctuations from the epoch of last scattering. The temperature anisotropy $\delta T/T$ is identified with the entropy fluctuation $\delta S/S$ at last scattering:

(12.5)

δ

$T/T|$

recombination

$\approx$

δ

S

q

$/S$

q

$|$

recombination

The predicted power spectrum of these fluctuations, computed from the Tsallis entropy fluctuation spectrum of the pre-recombination plasma, is:

(12.6)

$$P(k) \sim k^{n_s}$$

s

–

$$1) \cdot [1 + (q$$

–

$$1) \cdot k^2$$

ℓ

PI

$$^2 + O(q$$

–

$$1)^2]$$

where n_s is the standard scalar spectral index ($n_s \approx 0.965$ from Planck observations). The Tsallis correction term $(q-1) \cdot k^2 \ell_{Pl}^2$ is negligible at CMB scales ($k \sim 10^{-2} \text{ Mpc}^{-1}$) but becomes detectable at tensor-to-scalar ratios r — providing a further observational handle on the non-extensivity parameter q .

13. Philosophical Implications

13.1 Ontology of Entropicity

The Theory of Entropicity, if correct, implies a radical revision of our understanding of physical reality. The traditional ontology of physics takes matter-energy as the fundamental substrate of the universe: particles and fields exist in spacetime, carry energy, and interact through forces. In this ontology, entropy is a secondary quantity

— a measure of our ignorance about the precise microstate of a material system, or a measure of the disorder of an energy distribution.

The ontology of Entropicity reverses this hierarchy. In ToE:

- **Reality is made of information-entropy.** The fundamental ontological substrate of the universe is the entropic manifold $\mathbb{N}$, not particles, fields, or spacetime. Particles, fields, and spacetime are all projections, representations, or coarse-grainings of $\mathbb{N}$.
- **Matter-energy is derived.** The energy-momentum tensor $T_{\mu\nu}$ is a derived quantity, computed from the entropy gradients and information density of $\mathbb{N}$. Matter does not cause spacetime curvature; rather, concentrations of entropy (and the long-range entropic correlations they produce) generate what we perceive as matter and gravity.
- **The universe is self-computing.** The universe is a self-computing entropic system: it is its own computer, its own program, and its own output. The physical laws are the computation rules; the entropic manifold $\mathbb{N}$ is the computational medium; and the history of the universe is the computation's output.
- **Time is entropic.** Without an entropy gradient, there is no physical time. The second law of thermodynamics is not a contingent feature of our universe but a necessary consequence of the geometry of $\mathbb{N}$ — of the fact that the entropic manifold has a definite orientation (an "arrow") determined by the Obidi Action. Time exists because entropy changes; and entropy changes because the universe is a self-computing system that is not at its fixed point.

13.2 Mind, Consciousness, and the Entropic Universe

The No-Rush Theorem has significant implications for the philosophy of mind. If every computational process — including neural computation — is subject to the entropic speed limit (7.1), then consciousness, as a physical process, has a fundamental lower bound on the time required for any cognitive act. This does not

merely say that neurons are slow; it says that no physical process — regardless of the substrate, be it neurons, silicon, or any other material — can perform a cognitive operation faster than the entropic speed limit allows.

More broadly, consciousness in ToE is interpreted as a **high-order entropic feedback loop** within the universe's entropic manifold. A conscious system is one that maintains a persistent, self-referential sub-manifold of $\mathbb{N}$ — a region of $\mathbb{N}$ that models other regions of $\mathbb{N}$, including itself. The self-referentiality of consciousness is the physical instantiation of the Gödel incompleteness theorem on the entropic manifold: a sufficiently complex entropic sub-manifold cannot fully model its own entropic dynamics, just as a sufficiently powerful formal system cannot prove its own consistency.

Free will in entropic terms. The question of free will — whether agents have genuine causal power over their actions, or whether all actions are determined by prior causes — is reframed in ToE. An agent is a localized region of the entropic manifold that maintains coherent, low-entropy structure against the global entropic flow. Free will, in ToE, is the degree to which an agent's local entropic structure can resist, redirect, or deflect the global entropic flow — the extent to which local entropic computation can create a local decrease in entropy (an ordered, intentional act) by creating a larger increase in the entropy of the environment. In this sense, free will is physically real but always operates within the constraints of the Master Entropic Equation — it is entropic causation operating within the bounds of entropic law.

13.3 The End of the Universe

The classical picture of the end of the universe — Clausius's heat death [3] — is that the universe approaches a state of maximum entropy $S \rightarrow S_{\max}$, all temperature gradients vanish, all work is impossible, and time ceases. In ToE, this picture is refined and completed.

Heat death in ToE corresponds to the approach of the entropic feedback loop to its stable fixed point — the de Sitter spacetime with flat $\mathbb{N}$ and $\mathfrak{R}_\epsilon = 0$. At this fixed point, the entropy is at its maximum, the Fisher information tensor $T^{(\text{information})}_{\mu\nu}$ vanishes (no quantum correlations remain), and the Tsallis correction tensor $\Phi_{\mu\nu} \rightarrow 0$ (entropy is

maximally extensive). Time, in ToE, ceases — or more precisely, the rate of entropic change $dS/d\tau \rightarrow 0$, and the proper time τ becomes undefined because there are no physical processes to define its passage.

However, ToE suggests a possibility of **entropic resurrection**. Even at maximum entropy, the quantum vacuum has a non-zero Fisher information density $\rho_{\text{information}} \sim \hbar c / \mathcal{L}_{\text{de Sitter}}^4$, where $\mathcal{L}_{\text{de Sitter}} = c/\sqrt{\Lambda/3}$ is the de Sitter horizon radius. These quantum informational fluctuations are precisely the Boltzmann-brain fluctuations discussed by Penrose [11] and others: rare, enormously improbable spontaneous configurations of the vacuum that locally decrease entropy. In ToE, such fluctuations are described as instabilities of the de Sitter fixed point under the Fisher information term — precisely the same instability mechanism (but now at the de Sitter scale rather than the Planck scale) that produced the Big Bang at the beginning. The end of the universe, in ToE, is not a final death but a return to the condition that generated the universe in the first place: an unstable entropic fixed point, waiting for a quantum fluctuation to restart the entropic feedback loop.

14. Conclusion

We have presented the **Theory of Entropicity (ToE)** — a comprehensive, mathematically rigorous, and philosophically coherent unified theory of physics in which entropy is the fundamental ontological substrate of physical reality. The theory rests on three principal constructions:

The **Entropic Manifold** $\mathbb{N}$, a smooth Riemannian manifold equipped with the Fisher-Rao information metric, whose coordinates parametrize probability distributions over physical microstates, and whose geometry encodes the informational structure of spacetime. Physical spacetime is a coarse-grained projection of $\mathbb{N}$; quantum mechanics and general relativity are limiting regimes of the geometry of $\mathbb{N}$.

The **Obidi Action** — the master variational functional of ToE — defined on $\mathbb{N}$ and incorporating the entropic Ricci scalar, the matter Lagrangian, the Tsallis non-extensive entropy, the Fisher information of the metric, and the Von Neumann entropy density. All known action principles of physics (Einstein-Hilbert, Standard Model, quantum information theory, and Verlinde's entropic gravity) are limiting cases of the Obidi Action. The Obidi Action is diffeomorphism-invariant on $\mathbb{N}$, satisfies entropic general covariance, and provides a natural, UV-regulated, background-independent framework for all of physics.

The **Master Entropic Equation**, obtained by varying the Obidi Action with respect to the entropic metric, is a unified field equation that generalizes Einstein's equations to include entropy-stress and Fisher information tensors on the right-hand side, and Tsallis non-extensive corrections on the left. The Master Entropic Equation has three distinct regimes: the classical GR limit (Entropic Einstein Equation), the quantum limit (spectral dynamics of the entropic operator $\hat{E}$), and the non-extensive cosmological limit (Tsallis cosmology with dynamical dark energy).

The **No-Rush Theorem** establishes that the rate of entropic change along any physical trajectory is bounded above by the Margolus-Levitin entropic velocity. This theorem unifies the second law of thermodynamics, the quantum speed limit, and the emergence of coordinate time from entropic flow into a single geometric statement on $\mathbb{N}$. It implies that all physical processes — from quantum computation to neural cognition to cosmic expansion — are subject to a fundamental entropic speed limit.

The computational universe interpretation — extending Wheeler's "It from Bit" and Wolfram's Ruliad to the entropic domain — identifies the physical universe with a self-computing entropic system: the laws of nature are the geodesic equations and field equations of $\mathbb{N}$; the history of the universe is the output of the entropic computation; and observers are persistent, self-referential entropic sub-manifolds that sample coherent slices of the entropic superposition.

The Theory of Entropicity makes specific, quantitative, and experimentally testable predictions: a modified dispersion relation for photons detectable with FGST and CTA; a time-varying dark energy equation of state measurable by EUCLID and LSST; gravitational wave echoes detectable by LISA and Einstein Telescope; anomalous

heat capacity scaling in ultracold atomic gases; and precision tests of the Margolus-Levitin bound in superconducting quantum processors. These predictions distinguish ToE from standard GR + QFT and provide a clear experimental program for the coming decade.

Looking forward, the principal open problems are: (1) the full non-perturbative quantization of the Obidi Action via the entropic path integral (9.3), including computation of the S-matrix of entropic graviton-matter scattering; (2) the numerical simulation of the entropic feedback loop cosmology, including the full computation of the CMB power spectrum from Tsallis fluctuations; (3) the detailed derivation of the Standard Model parameters (coupling constants, particle masses, mixing angles) from the geometry of $\mathbb{N}$ and the values of q , α , and β ; and (4) the experimental search for Tsallis statistics and modified dispersion relations in precision quantum optical and cosmological experiments.

The Theory of Entropicity proposes that we live not in a universe of matter and energy, but in a universe of information and entropy — a universe that computes itself into existence through the ceaseless flow of entropy on the manifold $\mathbb{N}$, and whose laws of nature are, at bottom, the geometry of information.

Appendices

Appendix A: Derivation of the Fisher-Rao Metric as Entropic Spacetime Metric

We show that the Fisher-Rao metric $g^{(F)}_{ij}$ defined in (3.1) is the unique Riemannian metric on the space of probability distributions that is (i) invariant under sufficient statistics (Chentsov theorem), (ii) invariant under diffeomorphisms of the sample space X , and (iii) consistent with the positivity and normalization constraints on probability distributions.

Step 1: Reparametrization invariance. Let $\varphi: X \rightarrow X$ be a measurable bijection (sufficient statistic). Then $\varphi_*p(A) = p(\varphi^{-1}(A))$ defines a pushforward distribution. The Fisher metric satisfies $g^{(F)}_{ij}[p] = g^{(F)}_{ij}[\varphi_*p]$ for all sufficient statistics φ — this is Chentsov's theorem.

Step 2: Positivity. The Fisher-Rao metric is positive definite: for any tangent vector v^i at $p \in \mathbb{N}$, $g^{(F)}_{ij}v^iv^j = \int p(x|q)[v^i\partial_i \ln p]^2 dx \geq 0$, with equality if and only if $v = 0$.

Step 3: Identification with spacetime metric. In the GR limit, the probability distribution $p(x|q)$ is peaked at a single spacetime event $x_0(q)$: $p(x|q) \rightarrow \delta(x - x_0(q))$. In this limit:

g

(F)

ij

$(q) =$

$\int$

$p(x|q) [$

∂

i

$\ln p]^2 dx$

$\rightarrow$

g

$\mu\nu$

$(x$

0

) (

∂

x

0

μ

/

∂

q

i

) (

∂

x

0

v

/

∂

q

j

)

which is precisely the pullback of the spacetime metric $g_{\mu\nu}$ to the parameter space via the map $q \rightarrow x_0(q)$. This confirms that the Fisher-Rao metric reduces to the standard spacetime metric in the classical (delta-function) limit.

Appendix B: Tsallis Entropy and Non-Extensive Statistical Mechanics — Reference Formulas

Quantity	Tsallis ($q \neq 1$)	Boltzmann-Gibbs ($q \rightarrow 1$)
Entropy	$S_q = k_B(1 - \sum p_i^q)/(q-1)$	$S_{BG} = -k_B \sum p_i \ln p_i$
Equilibrium distribution	$p_i \propto [1 - (1-q)\beta E_i]^{1/(1-q)}$	$p_i \propto e^{-\beta E_i}$
Partition function	$Z_q = \sum [1 - (1-q)\beta E_i]^{1/(1-q)}$	$Z = \sum e^{-\beta E_i}$
Free energy	$F_q = -k_B T \ln_q Z_q$	$F = -k_B T \ln Z$
Additivity	$S_q(A+B) = S_q(A) + S_q(B) + (1-q)S_q(A)S_q(B)/k_B$	$S(A+B) = S(A) + S(B)$
Optimizing distribution	q-Gaussian: $e^{-x^2} \rightarrow [1 + (q-1)x^2]^{1/(q-1)}$	Gaussian: e^{-x^2}

where $\ln_q(x) = (x^{1-q} - 1)/(1-q)$ is the q -logarithm, satisfying $\lim_{q \rightarrow 1} \ln_q(x) = \ln(x)$.

Appendix C: Spectral Decomposition of the Entropic Operator $\hat{E}$

The Entropic Operator $\hat{E} = -k_B \text{Tr}(\rho \ln \rho)$ acts on the Banach space of trace-class operators on the Hilbert space $\mathcal{H}$. Its spectral theory proceeds as follows.

Spectral decomposition. For a finite-dimensional Hilbert space ($\dim \mathcal{H} = N < \infty$), the density matrix ρ has a complete spectral decomposition $\rho = \sum_{i=1}^N \lambda_i |\psi_i\rangle\langle\psi_i|$ with $\lambda_i \geq 0$ and $\sum \lambda_i = 1$. The Entropic Operator evaluates to $\hat{E}[\rho] = -k_B \sum \lambda_i \ln \lambda_i$, which is the Shannon entropy of the eigenvalue distribution $\{\lambda_i\}$.

Eigenstates of $\hat{E}$. Pure states $\rho = |\psi\rangle\langle\psi|$ are eigenstates of the "minimizers" of $\hat{E}$: they have $\hat{E}[|\psi\rangle\langle\psi|] = 0$ (zero entropy). The maximally mixed state $\rho = I/N$ has $\hat{E} = k_B \ln N$ (maximum entropy). The Entropic Operator is therefore a concave functional on the space of density matrices, with minimum 0 (pure states) and maximum $k_B \ln N$ (maximally mixed state).

Entropic Commutator. The commutation relation $[\hat{E}, \hat{H}] = i\hbar\hat{S}$ implies that $\hat{E}$ and $\hat{H}$ cannot be simultaneously diagonalized — they are complementary observables in the entropic sense. The entropy-flow operator $\hat{S}$ is the superoperator:

$$\hat{S}[\rho] = (i/\hbar)[\hat{H}, \rho] = - (1/\hbar)(d\hat{E}/dt)$$

which generates the entropic time evolution via the Lindblad-von Neumann equation.

Appendix D: Proof of the No-Rush Theorem (Full)

We provide the complete proof of the No-Rush Theorem, extending the sketch of Section 7.2.

Lemma 1 (Cramér-Rao bound on N). For any unbiased estimator of the entropy S from observations of the system, the variance satisfies $\text{Var}(\hat{S}) \geq 1/I_{\text{Fisher}}$, where $I_{\text{Fisher}} = g^{(F)}_{ij}(\nabla S)^i(\nabla S)^j$ is the Fisher information for the entropy parameter.

Lemma 2 (Bures metric bound). The speed of a trajectory $\gamma(\tau)$ on $\mathbb{N}$ in the Bures metric satisfies:

$$|d$$

γ

$/d$

τ

$|$

Bures

$\leq$

2

Δ

$E/\hbar$

where ΔE is the energy uncertainty of the state $\rho(\tau)$. This is the Anandan-Aharonov speed limit.

Theorem (No-Rush, full proof). Combining Lemma 1 and Lemma 2 via the Cauchy-Schwarz inequality:

$|dS/d$

τ

$| = |$

$\langle$

∇

S, d

γ

$/d$

τ

$\rangle$

F

$|$

$\leq$

$\parallel$

∇

$S\parallel$

F

$\cdot \parallel d$

γ

$/d$

τ

$\parallel$

F

$\leq$

$\sqrt{}$

$|$

Fisher

$\cdot (2$

Δ

$E/\hbar)$

By the Cauchy-Schwarz inequality for the Fisher inner product and the Margolus-Levitin bound $E \geq \hbar\pi/(2\tau_\perp)$ for the orthogonality time $\tau_\perp$:

$$\begin{aligned}
 & \left| \frac{dS}{dt} \right| \\
 & \leq \sqrt{\frac{2}{\pi}} \sqrt{\frac{E}{\hbar}} \sqrt{\tau_\perp} \\
 & \leq \sqrt{\frac{2}{\pi}} \sqrt{\frac{E}{\hbar}} \sqrt{\frac{\hbar\pi}{2E}} \\
 & = \frac{1}{\sqrt{2}}
 \end{aligned}$$

where the last inequality uses $\Delta E \leq \sqrt{(\pi k_B T E_{\text{system}})}$ for a thermal state at temperature T . This completes the proof. The bound is saturated for pure states evolving unitarily: $\rho(\tau) = |\psi(\tau)\rangle\langle\psi(\tau)|$, where $|d\gamma/d\tau|_{\text{Bures}} = \Delta E/\hbar$ exactly. $\square$

Appendix E: Comprehensive Comparison Table — ToE vs. Major Theories

Feature	GR	QFT	Verlinde	LQG	String Theory	ToE
Fundamental object	Metric $g_{\mu\nu}$	Quantum fields ϕ	Entropy on screens	Spin networks	Strings, branes	Entropic manifold $\mathcal{N}$
Entropy status	Derived	Derived	Partial foundation	Derived	Derived	Ontological primitive
Unification	No	No	Gravity only	Gravity + QM	All forces	All forces + gravity
Background independence	Yes	No	No	Yes	No	Yes
UV complete	No	No	No	Possibly	Possibly	Yes (Tsallis cutoff)
Dark matter	Unexplained	New particles	Partially	Unexplained	New particles	$T^{(\text{info})}_{\mu\nu}$ gradient
Dark energy	Fine-tuned Λ	Huge mismatch	Partially	Unexplained	Landscape	$\Lambda_\epsilon = S_q(\text{vac})/V$
Information paradox	Unresolved	Unresolved	N/A	Partial	Partial	Resolved
Free parameters	2 (G, Λ)	19	0 (derived)	1 (Barbero-Immirzi)	Many (landscape)	3 (q, α, β)
Experimental predictions	Many (confirmed)	Many (confirmed)	Few	Few	None distinct	Specific (GRB, CMB, GW)

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